MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§6.3

Adding and Subtracting Fractions

Unit fractions, estimating, and the bar, area, and number line models

Today's Goal

After completing this section, you should be able to:

- Add fractions with the same denominator by interpreting each fraction in terms of its unit fraction
- Estimate whether the sum of two fractions is less than or more than one whole before evaluating it
- Add and subtract fractions using the bar model, the area model, and the number line model, rewriting the fractions with a common denominator when needed

Learning Target 15

LT.15 Arithmetic with Fractions

I can apply standard strategies for addition, subtraction, multiplication, and division of fractions and rational numbers using appropriate representations.

Adding Quantities with Units

How would you figure out each of the following?

$2 \text{ days} + 4 \text{ days} = 6 \text{ days}$

$2 \text{ weeks} + 4 \text{ days} = 14 \text{ days} + 4 \text{ days} = 18 \text{ days}$

$3 \text{ feet} + 6 \text{ inches} = 36 \text{ inches} + 6 \text{ inches} = 42 \text{ inches}$

$1 \text{ hour} + 15 \text{ minutes} = 60 \text{ minutes} + 15 \text{ minutes} = 75 \text{ minutes}$

In order to add quantities with units, the units must match.

The Same Unit Fraction

With this in mind, we turn to our notion of thinking of fractions in terms of unit fractions. In order to add (or subtract) two fractions, they must have the same unit fraction.

Since the unit fraction associated with a fraction is determined by the fraction's denominator, this means that fractions need to have the same denominator in order for us to add or subtract them.

Evaluate Using Unit Fractions

Evaluate $\frac{4}{9} + \frac{3}{9}$ by interpreting each fraction in terms of the appropriate unit fraction.

$\frac{4}{9}$ is 4 one-ninths and $\frac{3}{9}$ is 3 one-ninths. Since both fractions are using the same unit fraction, we can simply add while keeping the same unit fraction.

$$\begin{array}{rrl} & 4 & \text{one-ninths} \\ + & 3 & \text{one-ninths} \\ \hline & 7 & \text{one-ninths} \end{array}$$

So $\frac{4}{9} + \frac{3}{9} = \frac{7}{9}$.

Adding with Different Denominators

Strictly speaking, fractions do not need to have the same denominator to be added or subtracted. What fraction does the bottom bar show?

$\frac{1}{3} =$ 

$\frac{1}{6} =$ 

$\frac{1}{3} + \frac{1}{6} =$ 

To interpret what fraction is being represented by that shaded region, we need to recognize that the red region is equivalent to two of the blue regions.

$\frac{3}{6} =$ 

Estimating

When adding (or subtracting) two fractions, it is often helpful to formulate an estimate of the result based on the size of the two fractions.

Without evaluating each expression, determine whether or not the sum should be less than one whole or more than one whole.

(a) $\frac{5}{8} + \frac{4}{7}$

Notice that $\frac{5}{8} > \frac{1}{2}$ and $\frac{4}{7} > \frac{1}{2}$. Since each fraction is more than $\frac{1}{2}$, adding them together would result in a sum that is more than one whole.

(b) $\frac{2}{5} + \frac{1}{3}$

Notice that $\frac{2}{5} < \frac{1}{2}$ and $\frac{1}{3} < \frac{1}{2}$. Since each fraction is less than $\frac{1}{2}$, adding them together would result in a sum that is less than one whole.

Is This Answer Correct?

Taking a moment to estimate can prevent us from making one of the most common mistakes students make when adding fractions.

A student was asked to evaluate $\frac{5}{8} + \frac{4}{7}$. The student's answer is shown below.

$$\frac{5}{8} + \frac{4}{7} = \frac{9}{15}$$

(a) Use estimation to explain why this answer is not correct.

Our estimate from Example 2 indicated that the sum should be more than one whole. But $\frac{9}{15}$ is less than one whole. So this student's answer cannot be correct.

Is This Answer Correct?

$$\frac{5}{8} + \frac{4}{7} = \frac{9}{15}$$

(b) What did the student do to get this answer?

The student added the numerators and added the denominators.

(c) Explain the conceptual error associated with this student's solution.

Adding denominators is a common mistake. Remember that the denominator represents the unit fraction being used. When adding fractions, the unit fractions associated with them are not added together.

Using the Bar Model

To add or subtract two fractions using the bar model, we begin by representing each fraction using the bar model. In doing so, make sure to use the same sized bar to represent one whole in each model.

Using the Bar Model

Use the bar model to evaluate $\frac{2}{7} + \frac{4}{7}$.

Note that both fractions are in terms of the unit fraction one-seventh. So we can divide each bar into 7 equal parts.

$\frac{2}{7} =$ 

$\frac{4}{7} =$ 

$\frac{2}{7} + \frac{4}{7} =$ 

So $\frac{2}{7} + \frac{4}{7} = \frac{6}{7}$.

Using the Bar Model

Use the bar model to evaluate $\frac{1}{2} + \frac{2}{5}$.

Since $\frac{1}{2}$ and $\frac{2}{5}$ have different denominators, we will begin by rewriting them as equivalent fractions with a common denominator. The least common denominator is 10: $\frac{1}{2} = \frac{1 \cdot 5}{2 \cdot 5} = \frac{5}{10}$ and $\frac{2}{5} = \frac{2 \cdot 2}{5 \cdot 2} = \frac{4}{10}$.

$\frac{5}{10} =$ 

$\frac{4}{10} =$ 

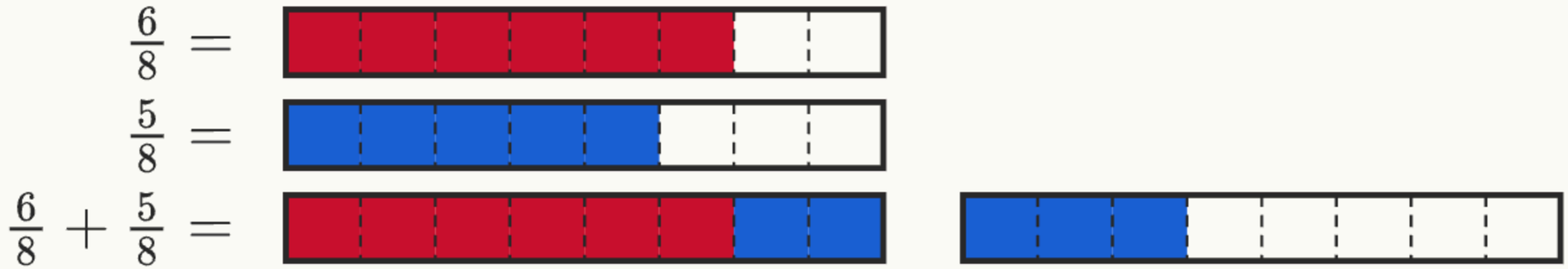
$\frac{5}{10} + \frac{4}{10} =$ 

So $\frac{1}{2} + \frac{2}{5} = \frac{9}{10}$.

Using the Bar Model

Use the bar model to evaluate $\frac{3}{4} + \frac{5}{8}$.

Since $\frac{3}{4}$ and $\frac{5}{8}$ have different denominators, we will begin by rewriting them as equivalent fractions with a common denominator. The least common denominator is 8, so $\frac{3}{4} = \frac{3 \cdot 2}{4 \cdot 2} = \frac{6}{8}$.



Note that the result is greater than one whole because we have one bar completely shaded and part of a second bar shaded. The model tells us that $\frac{3}{4} + \frac{5}{8} = \frac{11}{8}$.

Mixed Numbers

Note that the answer for Example 5 can also be written as a **mixed number**: $1\frac{3}{8}$, which is read as “one and three-eighths”.

This should not be confused with $1 \cdot \frac{3}{8}$, which means “one times three-eighths”.

Subtract Using the Bar Model

Work with a partner. Use the bar model to evaluate each difference.

(a) $\frac{5}{6} - \frac{1}{6}$ (b) $\frac{2}{3} - \frac{1}{4}$ (c) $\frac{3}{2} - \frac{3}{4}$

Make sure to use the same sized bar to represent one whole in each model.

How Many Parts Are Left?

(a) $\frac{5}{6} - \frac{1}{6}$

$\frac{5}{6}$ 

Take away 1 of the 5 red parts.

$\frac{5}{6} - \frac{1}{6}$ 

4 red parts are left, so $\frac{5}{6} - \frac{1}{6} = \frac{4}{6}$.

(b) $\frac{2}{3} - \frac{1}{4}$

$\frac{2}{3} = \frac{8}{12}$ and $\frac{1}{4} = \frac{3}{12}$.

$\frac{8}{12}$ 

Take away 3 of the 8 red parts.

$\frac{8}{12} - \frac{3}{12}$ 

5 red parts are left, so $\frac{2}{3} - \frac{1}{4} = \frac{5}{12}$.

How Many Parts Are Left?

(c) $\frac{3}{2} - \frac{3}{4}$

$\frac{3}{2} = \frac{6}{4}$, more than one whole, so we need two bars.

$\frac{6}{4}$ 

Take away 3 of the 6 red parts.

$\frac{6}{4} - \frac{3}{4}$ 

3 red parts are left, so $\frac{3}{2} - \frac{3}{4} = \frac{3}{4}$.

Using the Area Model

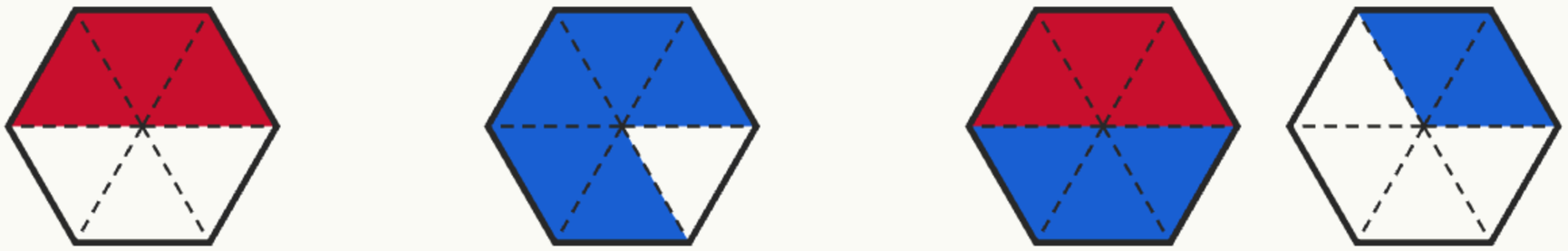
Remember that with the area model, we can use any convenient geometric shape to represent one whole. The usual choices are rectangles, squares, and circles. As with the bar model, it is important to use the same shape (and size) to represent one whole.

Using the Area Model

Use the area model to evaluate $\frac{1}{2} + \frac{5}{6}$.

The least common denominator is 6, so $\frac{1}{2} = \frac{1 \cdot 3}{2 \cdot 3} = \frac{3}{6}$. We will use a hexagon to represent one whole and divide each hexagon into six equal parts.

$\frac{3}{6}$ $+$ $\frac{5}{6}$ $=$ $\frac{8}{6}$



So $\frac{1}{2} + \frac{5}{6} = \frac{8}{6}$, which we can rewrite as a mixed number: $1\frac{2}{6}$ or, equivalently, $1\frac{1}{3}$.

Using the Number Line Model

We can use the number line model to represent addition and subtraction with fractions (and rational numbers) in the same way that we used it to represent addition and subtraction with integers.

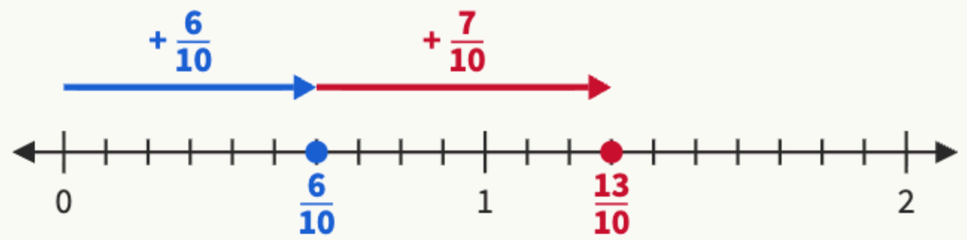
Using the Number Line Model

Use the number line model to evaluate $\frac{3}{5} + \frac{7}{10}$.

We will rewrite $\frac{3}{5}$ as an equivalent fraction, $\frac{3 \cdot 2}{5 \cdot 2} = \frac{6}{10}$. With both fractions in terms of one-tenths, this prompts us to divide each unit length on the number line into ten equal parts.

We begin with $\frac{6}{10}$ on the number line.

Then, to add $\frac{7}{10}$, we continue moving forward to the right a distance of $\frac{7}{10}$.



So $\frac{3}{5} + \frac{7}{10} = \frac{13}{10}$, which we can rewrite as a mixed number: $1\frac{3}{10}$.

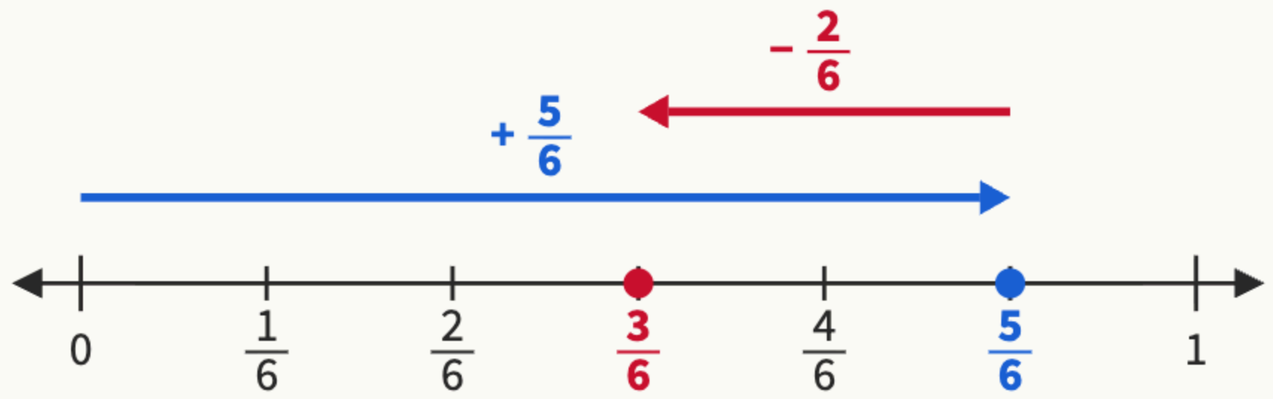
Using the Number Line Model

Use the number line model to evaluate $\frac{5}{6} - \frac{1}{3}$.

We rewrite $\frac{1}{3}$ as $\frac{1 \cdot 2}{3 \cdot 2} = \frac{2}{6}$.

We begin with $\frac{5}{6}$ on the number line.

Then, to subtract $\frac{2}{6}$, we turn around and move forward to the left a distance of $\frac{2}{6}$.



So $\frac{5}{6} - \frac{1}{3} = \frac{3}{6}$.

Find Two Fractions

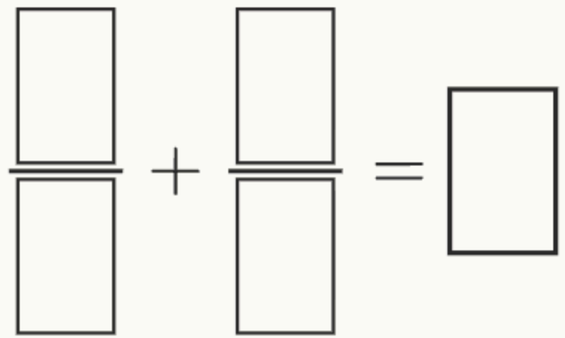
Use the digits 1 to 9, each at most one time, to make two fractions whose difference is $\frac{1}{2}$. Use any model to show that your answer is correct. How many different pairs can your group find?

A pair counts when the difference is exactly $\frac{1}{2}$ and a model shows it.

There are 23 pairs in all. Did another group find a pair that your group missed?

A Whole Number Sum

Fill in each of the boxes below with a digit from 1 to 9 (at most one time each) to create two fractions whose sum is a whole number. How many different whole numbers can you create?



There are 8 different whole numbers in all. Did another group find one that your group missed?